

Physical Models of Living Systems

Ecology, neural systems and brain criticality — Lecture Notes

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1. About these notes

1.1. Scope

These are lecture notes for *Physical Models of Living Systems* (third-semester elective, Master in Physics of Data, University of Padova, instructor Prof. Samir Suweis). The course uses the vocabulary of statistical physics — master equations, mean-field limits, phase transitions, random matrices, criticality — to model biological systems at scales that span from a single microbial population to a whole brain.

The notes are split into two halves that mirror the course:

- **Ecology.** How to write down and analyse a model of one, few, or many species living, reproducing and interacting. Deterministic and stochastic descriptions of population growth; neutral theory and macroecological patterns; the Lotka–Volterra family; random-matrix results on the complexity–stability trade-off.
- **Neural systems.** Firing-rate models for populations of neurons; Hebbian plasticity and associative memory (Hopfield networks); the Boltzmann machine as a probabilistic recurrent network; and the hypothesis that healthy brain dynamics live near a critical point.

1.2. Reading the notes

Every chapter is self-contained enough that it can be read on its own, but the two halves share a common toolbox: master equations and the Fokker–Planck / Langevin limit appear in ecology (Chapter 2) and again in neural dynamics (Chapter 5–7); the notion of a phase transition and of scaling laws frames both the neutral theory of biodiversity (Chapter 3) and brain criticality (Chapter 7).

Definitions, theorems, worked examples and remarks are collected in styled callouts so the key results can be scanned without reading through the derivations.

1.3. Notation

- $n(t), N(t)$: population size (single species / community total).
- $\rho(\mathbf{x}, t)$: probability density on state space; $P_n(t)$ its discrete counterpart.
- $\mathcal{L}$: master-equation generator; $D(x)$ and $A(x)$ drift and diffusion in Fokker–Planck.
- r_i, K_i, α_{ij} : intrinsic growth rate, carrying capacity, and pairwise interaction coefficient in Lotka–Volterra.
- $\phi_i(t)$: firing rate of neuron / population i ; W_{ij} synaptic weight.
- $\mathbf{s} = (s_1, \dots, s_N)$: Boolean neural state ($s_i \in \{0, 1\}$ or $\{\pm 1\}$).
- $H[\mathbf{s}]$: energy function of a Hopfield / Boltzmann network; $\beta = 1/T$ inverse temperature.

1.4. Sources

These notes are a re-organised and expanded version of *Physical Model of Living System* by Prof. Samir Suweis (Course Notes final, 2022–2023, written by G. Cataldi, F. Manzali, L. Rosset). Standard references

1. About these notes

consulted throughout are:

- Marc Mézard & Andrea Montanari, *Information, Physics, and Computation*, Oxford, 2009.
- Peter Dayan & Larry Abbott, *Theoretical Neuroscience*, MIT Press, 2005.
- Simon A. Levin (ed.), *The Princeton Guide to Ecology*, Princeton, 2009.
- Igor Volkov, Jayanth R. Banavar, Stephen P. Hubbell & Amos Maritan, *Neutral theory and relative species abundance in ecology*, *Nature* **424**, 1035 (2003).
- John Hertz, Anders Krogh & Richard Palmer, *Introduction to the Theory of Neural Computation*, Westview, 1991.

Each chapter closes with its own *Sources* section naming the specific lectures / book chapters from which its content was drawn.

Part I.

Part I: Ecology

2. Introduction

This chapter follows the opening of Prof. Suweis's notes. It uses two motivating examples — the collapse of the Atlantic cod fishery and the geographic distribution of malaria mortality — to introduce the vocabulary the rest of the course will rely on: *emergent property*, *critical transition*, *power law*, and *universality*.

2.1. Two motivating examples

Example 2.1 (*Cod fishing in the North Atlantic*)

The record of cod catches in the Atlantic Ocean over the twentieth century shows regular oscillations that track reproduction cycles, plus one enormous peak in the mid-1960s driven by industrial fishing. That peak is followed by a smaller local maximum and then a sharp collapse in the early 1990s — the population never recovered. The abruptness of the collapse (rather than the smooth decrease a naive linear model would predict) is what makes this a **critical transition**. Explaining such abrupt changes is one of the main threads of the course.

Example 2.2 (*Global distribution of malaria mortality*)

A worldwide map of deaths caused by malaria shows a strongly non-uniform, patchy geography. Two possible statistical models describe such spatial data. A **Gaussian** model captures approximately uniform distributions with concentrated variance. A **power-law** model captures strongly heterogeneous distributions with fat tails. Understanding which parameters push a population toward one shape or the other is a recurring question in ecology.

2.2. Emergent ecological patterns

Definition 2.1 (Emergent property). An **emergent property** of an ecosystem is a universal feature of the system as a whole — not deducible from any single species or component — that arises as a collective behaviour of the entire structure. When the same pattern is observed for many taxa across many continents it typically signals a **universal law** underlying the property.

Three empirical patterns dominate the field:

Definition 2.2 (Species–area relationship (SAR)). The **species–area relationship** is the empirical scaling $S(A) \propto A^z$, where $S(A)$ is the number of species observed in an area A and z is a dimensionless exponent. The relationship holds across taxa (amphibians, birds, mammals) and across continents; only the multiplicative prefactor changes, not z .

Definition 2.3 (Endemic–area relationship (EAR)). The **endemic–area relationship** is the analogous

2. Introduction

scaling $E(A) \propto A^{z'}$ for the number of *endemic* species — those found only in the sampled area A . Its exponent z' is close to $1 - z$ in most datasets.

Definition 2.4 (Relative species abundance (RSA)). The **relative species abundance** is the empirical distribution of species abundances in a community. Preston's log-normal and Fisher's log-series are the two standard fits; either shape is remarkably robust across ecosystems.

Remark. The universality of SAR, EAR and RSA across ecosystems is one of the empirical strongholds of theoretical ecology. It is the reason why *neutral theory* (Chapter 3) — which reproduces all three patterns from a single mechanism — was received as a paradigm shift when it was proposed in the early 2000s.

2.3. Scope of the course

The scientific goal is threefold: (i) understand why regular patterns emerge in systems with heterogeneous components and noisy dynamics; (ii) identify the *fundamental* mechanisms responsible; (iii) predict how the patterns respond to perturbations (climate, extinction, disease).

The methodological answer is the toolbox of statistical physics — **master equations**, mean-field limits, phase transitions, random matrices, criticality — applied to biological systems. Chapters 2–4 build the ecology half of the toolbox, and Chapters 5–10 build the neural systems half.

Proposition 2.1 (When statistical physics applies to biology). *A physics-of-biology model is likely to be predictive when the system satisfies three conditions:*

1. *many degrees of freedom ($N \gg 1$),*
2. *local, sparse, or mean-field interactions,*
3. *some form of noise (thermal, demographic, synaptic) that makes ensemble averages well defined.*

2.4. Critical transitions

The cod fishery collapse is one instance of a general phenomenon: some systems undergo qualitative changes in their macroscopic behaviour as a control parameter is varied, and the change can be discontinuous.

Definition 2.5 (Critical transition). A **critical transition** is a qualitative change in the macroscopic behaviour of a system that occurs at a critical value of some control parameter. At the transition point observables (order parameters, correlation lengths, response functions) show universal scaling, and small perturbations can have disproportionately large effects.

Remark (Physical meaning). Critical transitions in ecology (fishery collapse, forest–savanna shift, coral bleaching) mirror those in physics (magnetisation, superfluid transition). This parallel is not merely rhetorical: near a critical point the systems really are in the same *universality class*, share the same critical exponents, and can be analysed with the same renormalisation-group tools.

2.5. Book supplements

The single most useful supplementary reference is Volkov, Banavar, Hubbell & Maritan, *Neutral theory and relative species abundance in ecology*, Nature **424**, 1035 (2003), which lays out all three empirical patterns and shows how one mechanism explains them. On critical transitions in ecology, Marten Scheffer's *Critical Transitions in Nature and Society* (Princeton, 2009) is a comprehensive treatment.

2.6. Exercises

Exercise 2.1

Verify that a SAR of the form $S(A) \propto A^z$ implies a species–individual relationship $S(N) \propto N^z$ if abundance is proportional to area. State the assumptions.

Solution

If $N \propto A$ (density is spatially constant), then substituting $A \propto N$ into $S \propto A^z$ gives $S \propto N^z$ with the *same* exponent. The result relies on the assumption of spatial homogeneity of the abundance density; when that fails (as in strongly clumped populations) the two exponents diverge.

Exercise 2.2

For the Fisher log-series RSA, the number of species with k individuals is $\phi_k = \theta x^k / k$ with parameters $\theta, x < 1$. Show that the total number of species is $S = -\theta \ln(1-x)$ and the total number of individuals is $J = \theta x / (1-x)$.

Solution

$S = \sum_{k \geq 1} \theta x^k / k = -\theta \ln(1-x)$ (Taylor series of $\ln$). $J = \sum_{k \geq 1} k \phi_k = \theta \sum_{k \geq 1} x^k = \theta x / (1-x)$ (geometric series). Eliminating x recovers the *fundamental biodiversity number* $\theta = J / \ln(J/S)$ for large J/S .

2.7. Callouts

Note

The introduction is largely *phenomenological*. Definitions of SAR, EAR, RSA are made precise here so the mathematical developments in Chapters 3–4 can be tested against them.

Tip

Whenever you see a scaling law in the field, check its *exponent* first: universal exponents point to universal mechanisms; system-specific exponents point to system-specific biology. This is the physicist's first triage step.

Warning

The cod-fishery example is a paradigm of a **regime shift**, not a smooth decline. Statistical tests for regime shifts (Cusum, Bayesian change-point) differ from tests for gradual trends. Applying trend tests to shift data misses the phenomenon.

2.8. Sources

Prof. Suweis, *PMLS Course Notes*, Preface + Chapter 1. Volkov, Banavar, Hubbell & Maritan (2003) *Neutral theory and relative species abundance in ecology*, *Nature* **424**, 1035. Scheffer (2009), *Critical Transitions in Nature and Society*, Princeton.

3. Dynamics of a single species

This chapter follows Prof. Suweis's Chapter 2. It builds from the simplest ecological model — a single bacterial population — up to a stochastic description that captures demographic noise. The four subsections mirror the source: (i) exponential and logistic growth from cell-division kinetics; (ii) the consumer–resource model and the quasi-stationary approximation; (iii) the stochastic birth–death master equation and its Fokker–Planck / Langevin limits; (iv) the stationary distribution and stochastic doubling time.

3.1. 2.1 Population dynamics: exponential and logistic growth

The paradigmatic microbial system is *E. coli* in a chemostat. Under the microscope a single bacterium grows up to a critical size and then divides into two — a **cascade reproduction** process. In optimal conditions the reproduction time is $\tau \approx 20$ minutes.

Definition 3.1 (Exponential growth). Let $N(t)$ be the population size at time t . If per-capita reproduction happens at rate μ , the deterministic mean-field law is

$$\frac{dN}{dt} = \mu N \implies N(t) = N(0) e^{\mu t} = N(0) 2^{t/T},$$

with **doubling time** $T = \ln 2 / \mu$. For *E. coli* in optimal conditions $\mu \approx 2.1 \text{ h}^{-1}$, so $T \approx 20$ min.

Remark. Two implicit assumptions: (i) N is treated as a continuous variable, which is a poor approximation for $N \sim O(1)$ near $t = 0$; (ii) resources are assumed unlimited. Removing (ii) is what motivates the logistic model below.

The resource-limited fix is Verhulst's:

Definition 3.2 (Logistic growth). With **carrying capacity** $k > 0$ the density-dependent update is

$$\frac{dN}{dt} = \mu N \left(1 - \frac{N}{k} \right),$$

whose explicit solution is

$$N(t) = \frac{N(0) e^{\mu t}}{1 + \frac{N(0)}{k} (e^{\mu t} - 1)}.$$

The two fixed points are $N^* = 0$ (unstable if $\mu > 0$) and $N^* = k$ (stable).

Example 3.1 (*Bacterial growth in a chemostat*)

A batch culture of *E. coli* fits the logistic law until nutrients deplete. The lag phase corresponds to $N \ll k$ where $N(t) \approx N(0) e^{\mu t}$; the plateau corresponds to $N \rightarrow k$. Fitting μ and k from an OD-600 curve is one of the first assays a biophysics student runs.

3.1.1. 2.1.1 Consumer Resource Model

Rather than encoding resource limitation in an ad-hoc carrying capacity, we can model the resource $R(t)$ explicitly. Let $g(R)$ be the resource uptake rate per unit cell and $\mu(R)$ the growth-resource function. With uptake efficiency c , yield coefficient γ (number of nutrient units to produce a bacterium), and death rate d :

Definition 3.3 (Consumer–resource dynamics).

$$\frac{dN}{dt} = (\gamma c g(R) - d) N, \quad \frac{dR}{dt} = \mu(R) - c g(R) N.$$

To get a closed-form population equation, assume the resource equilibrates much faster than the consumers (the **quasi-stationary approximation, QSA**): set $dR/dt = 0$ and eliminate R . For a biotic resource with logistic dynamics of its own, $\mu(R) = Rr^*(1 - R/k_c)$, algebra gives an *effective* logistic law

$$\frac{dN}{dt} = N(\tilde{k} - \tilde{a}N),$$

with effective growth rate $\tilde{k}$ and effective saturation $\tilde{a}$ built from the microscopic constants $\{s, \mu, c, e, m\}$.

Proposition 3.1 (Effective logistic from CRM). *In the quasi-stationary limit of the consumer–resource model the population dynamics reduce to*

$$\dot{N} = r N(1 - N/K) + O(N^2),$$

with r and K explicit combinations of the microscopic rates. The logistic law of §2.1 is thus a derived description, not a phenomenological guess.

3.2. 2.2 Stochastic modelling of population growth

The deterministic ODE ignores the discreteness of individuals and the demographic noise that comes with it. Both matter when N is small.

3.2.1. 2.2.1 Birth–death master equation

Definition 3.4 (Master equation for a birth–death process). Let $P_n(t)$ be the probability the population is n at time t , with per-cell birth rate b_n and death rate d_n . The **master equation** is

$$\frac{dP_n(t)}{dt} = b_{n-1}P_{n-1}(t) + d_{n+1}P_{n+1}(t) - (b_n + d_n)P_n(t).$$

The right-hand side is *gain minus loss* of probability at state n .

For the *linear* case $b_n = bn$, $d_n = dn$, the moments obey closed equations.

Theorem 3.2 (Mean and variance of a linear birth–death process). *Starting from $n(0) = n_0$,*

$$\langle n(t) \rangle = n_0 e^{(b-d)t}, \quad \text{Var}[n(t)] = n_0 \frac{b+d}{b-d} (e^{2(b-d)t} - e^{(b-d)t}).$$

The Fano factor $F = \text{Var}[n]/\langle n \rangle$ grows without bound whenever $b > d$: fluctuations are strongly super-

Poissonian.

3.2.2. 2.2.2 Fokker–Planck and Langevin equations

Rescaling $n = \Omega x$ with $\Omega \gg 1$ and expanding the master equation to second order gives the *system-size expansion* (Kramers–Moyal):

Theorem 3.3 (Fokker–Planck limit). *In the limit of large system size Ω ,*

$$\partial_t \rho(x, t) = -\partial_x [A(x)\rho] + \frac{1}{2\Omega} \partial_x^2 [B(x)\rho],$$

*with drift $A(x) = \lim_{\Omega \rightarrow \infty} \Omega^{-1}(b_{\Omega x} - d_{\Omega x})$ and diffusion $B(x) = \lim_{\Omega \rightarrow \infty} \Omega^{-1}(b_{\Omega x} + d_{\Omega x})$. The corresponding (Itô) SDE — the **Langevin equation** — is*

$$dx = A(x) dt + \sqrt{B(x)/\Omega} dW_t.$$

For the logistic birth–death process ($b_n = bn$, $d_n = dn + qn^2/K$) one finds $A(x) = rx(1 - x)$ and $B(x) = (b + d)x$.

3.2.3. 2.2.3 Random walk

Close to the extinction boundary $x \approx 0$, the drift is $A \approx rx$ and the diffusion $B \approx (b + d)x$, so

$$dx \approx rx dt + \sqrt{(b + d)x/\Omega} dW_t.$$

A Lamperti-type transform $y = 2\sqrt{x\Omega/(b + d)}$ maps this to a *biased random walk*:

$$dy = \tilde{r} dt + dW_t.$$

Example 3.2 (Extinction of a critical branching process)

Setting $r = 0$ ($b = d$) makes the walker unbiased. Standard random-walk theory shows that a walker on the half-line with a reflecting or absorbing barrier at 0 hits 0 with probability one: the population is extinct almost surely, even though its mean is conserved.

3.2.4. 2.2.4 Stationary solution with linear rates

For a birth–death process satisfying detailed balance the stationary distribution is a product of rate ratios:

$$P_n^{ss} = P_0^{ss} \prod_{k=1}^n \frac{b_{k-1}}{d_k}.$$

Proposition 3.4 (Stationary of logistic birth–death). *For $b_n = bn$ and $d_n = dn + qn^2/K$ the stationary distribution has mode near the deterministic fixed point $n^* = K(1 - d/b)$, with variance $O(K)$. In the deterministic limit $K \rightarrow \infty$ the distribution concentrates at n^* .*

3.2.5. 2.2.5 Stochastic doubling

For a linear birth–death process with $b > d$ the **mean** doubling time is

$$\tau_2 = \ln 2 / (b - d),$$

independent of n_0 . The *fluctuations* have coefficient of variation

$$\frac{\sigma_{\tau_2}}{\tau_2} \sim \sqrt{\frac{b + d}{n_0(b - d)}},$$

so relative fluctuations matter only for small populations. This is why single-cell lineage-tracking experiments see a broad distribution of division times.

Remark (Physical meaning). The three levels of description — master equation, Fokker–Planck, Langevin — are the same process at different resolutions: the master equation is exact but discrete; the Fokker–Planck retains fluctuations of order $\Omega^{-1/2}$; the deterministic ODE $\dot{x} = A(x)$ keeps only the drift. Going up the ladder trades tractability for exactness.

3.3. Exercises

Exercise 3.1

Solve the logistic ODE explicitly. Verify that $N(t) \rightarrow K$ as $t \rightarrow \infty$ for any $N_0 > 0$.

Solution

Separate variables:

$$\frac{dN}{N(1 - N/K)} = \mu dt.$$

Partial-fraction expand the left-hand side, integrate, and exponentiate to obtain

$$N(t) = \frac{K}{1 + (K/N_0 - 1)e^{-\mu t}}.$$

For $\mu > 0$ the exponential decays, so $N(t) \rightarrow K$.

Exercise 3.2

Show that the linear birth–death process with $b = d$ has $\langle n(t) \rangle = n_0$ but $\text{Var}[n(t)] = 2bn_0t$. Interpret the ratio $\sigma/\langle n \rangle$.

Solution

Set $b = d$ in the mean/variance formulas and take the limit $b - d \rightarrow 0$ using l'Hôpital. Variance grows linearly in time: the Fano factor grows as $2bt$, so relative fluctuations grow without bound. A critical branching process eventually goes extinct with probability one *even though its mean is conserved*.

Exercise 3.3

Derive the QSA effective logistic law from the biotic consumer–resource system. Identify r and K explicitly in terms of $\{r^*, k_c, c, \gamma, d\}$.

Solution

Setting $\dot{R} = 0$: $R^* = k_c(1 - cN/r^*)$. Substituting into $\dot{N}/N = \gamma c R^* - d$ gives $\dot{N} = N(\tilde{k} - \tilde{a}N)$ with $\tilde{k} = \gamma c k_c - d$ and $\tilde{a} = \gamma c^2 k_c / r^*$. Matching with the standard $\dot{N} = rN(1 - N/K)$ yields $r = \tilde{k}$ and $K = \tilde{k}/\tilde{a} = r^*(\gamma c k_c - d)/(\gamma c^2 k_c) = r^*(1 - d/(\gamma c k_c))/c$.

3.4. Callouts

i Note

The logistic law is a phenomenological summary of many microscopic mechanisms (nutrient limitation, quorum sensing, waste build-up). Fitting a curve to it does not identify a mechanism.

! Important

Whenever a deterministic ODE has a fixed point at $n^* = 0$ and a birth term proportional to n , extinction has finite probability at all finite n_0 . Deterministic stability is not the same as stochastic survival.

💡 Tip

When simulating a stochastic birth–death process, use the Gillespie algorithm rather than a discretised Langevin scheme for small n : the Gillespie algorithm samples the exact interarrival distribution and preserves integer n .

⚠ Warning

The Fokker–Planck limit assumes $\Omega \gg 1$: it fails near extinction ($n \ll \Omega$). If you care about extinction probabilities, go back to the master equation.

🔥 Caution

Fitting parameters (r, K) from data by nonlinear least squares often looks fine but hides large uncertainties: the correlation between r and K in the logistic fit is close to ± 1 unless the early exponential phase is well-sampled.

3.5. Book supplements

The system-size expansion and the birth–death formalism are treated in canonical detail in van Kampen, *Stochastic Processes in Physics and Chemistry* (Elsevier, 2007), Chapters IV, VII and XI. The QSA and the consumer–resource derivation are covered in Nowak, *Evolutionary Dynamics* (Harvard, 2006), Chapter 3.

3.6. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 2. van Kampen (2007), Chapters IV, VII, XI. Nowak (2006), Chapter 3.

4. Multiple species and neutral theory

Real ecosystems host many species at once, and a lot of ecology consists in trying to explain the *macroecological patterns* — species-abundance distributions, species–area relationships, α -diversity across sites — that are seen empirically in almost every biome. This chapter introduces the classical *neutral theory* of Hubbell, in which those patterns are generated without any biological interaction: only birth, death and immigration acting on ecologically equivalent species.

4.1. Motivation: from bacteria to forests

A striking empirical fact is that the shape of the species-abundance distribution (SAD) — the histogram of the number of species with k individuals in a community — is remarkably similar across systems as different as tropical forests, coral reefs, and microbial communities. This suggests either (i) a fine-tuning of biological parameters across all ecosystems (unlikely), or (ii) an emergent statistical mechanism that is largely insensitive to the details.

Definition 4.1 (Neutral hypothesis). An ecological community satisfies the **neutral hypothesis** when all species share identical per-capita birth, death, and dispersal rates. Individuals are drawn from a common label pool but are otherwise interchangeable.

Remark (Physical meaning). The neutral hypothesis is a null model in the same sense that an ideal gas is a null model for a fluid. Every deviation from neutral predictions is a signature of biology (differential fitness, niche structure, competition).

4.2. The Hubbell model

Fix a community of exactly J individuals. At every time step: one individual dies at random, and it is replaced by (a) a copy of another individual drawn uniformly with probability $1 - m$, or (b) a fresh migrant drawn from an external *metacommunity* with probability m . This is a *zero-sum* Moran-like process with immigration.

Proposition 4.1 (SAD of the local neutral community). *In the neutral zero-sum model with metacommunity biodiversity number θ and immigration rate m , the expected number of species with k individuals in the local community of size J is given by*

$$\langle S_k \rangle = \theta \frac{J! \Gamma(k+I) \Gamma(J-k+I(1-m^{-1}))}{k! (J-k)! \Gamma(I) \Gamma(J+I)} \cdot \frac{\Gamma(\theta+I-\theta)}{\Gamma(\theta)},$$

where $I = m(J-1)/(1-m)$. The functional form is a zero-sum multinomial whose limiting shape for $m \rightarrow 0$ is Fisher's log-series, and for $m \rightarrow 1$ is a truncated log-series with heavier abundant-tail.

The result is somewhat unwieldy but its message is clean: two adjustable parameters (θ, m) are enough to reproduce SADs across many empirical datasets. That is a strong argument for neutral dynamics as an

4. Multiple species and neutral theory

organising principle, though not for its literal truth.

Example 4.1 (*Barro Colorado Island*)

The 50-hectare forest plot on Barro Colorado Island (Panama) contains $\sim 21,000$ trees of ~ 300 species. Fitting the Hubbell SAD to the observed abundance histogram gives $\theta \approx 50$, $m \approx 0.09$; the fit is essentially indistinguishable from a much more detailed niche-based model. This is the empirical result that made neutral theory famous.

4.3. Scaling and criticality

The neutral community sits at a *critical* point in the parameter space: fluctuations in species abundances are power-law-tailed, and the correlation length (the typical range over which two individuals are related) diverges as $m \rightarrow 0$. The physicist's diagnostic is that observables scale as power laws in system size.

Definition 4.2 (Scaling law). A quantity Q scales with system size L if $Q(L) \sim L^\gamma$ for some exponent γ in the asymptotic regime $L \rightarrow \infty$. Different observables sharing a small number of exponents that satisfy consistency relations define a **universality class**.

Theorem 4.2 (Species–area law). Under neutral dynamics on a lattice with dispersal length ℓ , the number of species found in an area A scales as

$$S(A) \sim A^z, \quad z \approx 0.25 \pm 0.1,$$

independent of the microscopic diffusion. The exponent z is a critical exponent — it depends on spatial dimensionality and the tail of the dispersal kernel, not on the specific species.

Corollary 4.3 (Endemic-area relationship). The number of species endemic to (i.e. found only in) an area A scales as $E(A) \sim A^{z'}$ with $z' \sim 1 - z$. This is the theoretical prediction that lets conservation biologists estimate how many species would be lost if a patch of habitat of area A were destroyed.

4.4. Phenomenological renormalisation group

A powerful test that a system is at a critical point is that its macroscopic distribution is *scale invariant*: coarse-graining leaves the shape of the SAD unchanged, only rescaled. The phenomenological RG procedure of Suweis et al. (2020) is to (i) partition a large empirical community into b sub-communities of equal size, (ii) compute the SAD of a randomly chosen sub-community, (iii) rescale abundances by $\langle n \rangle$ and species counts by $\langle S \rangle$. If the resulting distributions collapse across values of b , the community is scale-invariant — as is observed for tropical forests and reef fish.

Lemma 4.4 (Scale invariance implies power-law SAD). If the SAD $P(n)$ collapses under the coarse-graining $n \rightarrow n/b$, $P \rightarrow bP$, then it is a power law: $P(n) \sim n^{-\alpha}$ with α determined by the way $\langle n \rangle$ scales with b .

4.5. Inferring biodiversity from presence–absence data

In many ecological surveys the raw datum is not abundance but *presence–absence*: for each of N sites and each of S species, one bit says whether the species was seen at the site. Recovering the full SAD from presence–absence is a classical inference problem.

Proposition 4.5 (Presence–absence to SAD). *Under the neutral assumption, the probability that species σ is present at a site of size J given it has abundance n_σ in the metacommunity is*

$$p_\sigma^{\text{pres}} = 1 - (1 - n_\sigma/J_M)^J.$$

Inverting the relation with a beta prior on n_σ/J_M recovers the underlying θ from presence–absence alone. This is how tropical biodiversity is estimated from herbarium records.

4.6. Local dynamics and migration

Neutral theory can be extended to a *spatial* community, where individuals disperse over a distance drawn from a kernel $K(\mathbf{x})$. The kinetic equation for the density $\rho_\sigma(\mathbf{x}, t)$ of species σ is a spatial birth–death–diffusion process. In the mean-field limit,

$$\partial_t \rho_\sigma = D \nabla^2 \rho_\sigma + (b - d) \rho_\sigma - c \rho_\sigma \sum_{\sigma'} \rho_{\sigma'} + m \bar{\rho}_\sigma,$$

with the last term representing migration from the metacommunity mean $\bar{\rho}_\sigma$.

Example 4.2 (Species persistence and characteristic method)

For a single species with $b > d$, transformed variables allow the extinction time T_{ext} to be computed as a first-passage problem for a biased random walk. In the large- K limit,

$$\langle T_{\text{ext}} \rangle \sim K \exp(2K(b - d)/(b + d)),$$

i.e. *exponential* in the carrying capacity — extinction is rare but not impossible.

4.7. Grand canonical formulation

Just as in statistical mechanics one goes from the microcanonical to the canonical to the grand-canonical ensemble by relaxing conservation laws, neutral theory can be reformulated with a fluctuating total community size J . The grand-canonical partition function admits a saddle-point evaluation and reproduces all the closed-form results above from a single generating functional.

Remark. The grand-canonical route is *not* mere formalism: it is what lets you compare SADs across datasets of different total abundance, and to define intensive quantities (diversity per unit area, species-per-individual ratio) that are the actual observables reported in the field.

4.8. Exercises

Exercise 4.1

The metacommunity in the Hubbell model has S_M species and total abundance J_M . Show that the fundamental biodiversity number $\theta = J_M / \ln(J_M / S_M)$ in the limit $J_M \rightarrow \infty$ with $S_M \rightarrow \infty$ fixed by the ratio.

Solution

Assume the metacommunity SAD is a Fisher log-series with parameter θ : $\langle S_k \rangle = \theta x^k / k$ where x is chosen so that $\sum_k k \langle S_k \rangle = J_M$. Sum the geometric series to obtain $J_M = \theta x / (1 - x)$ and total species count $S_M = -\theta \ln(1 - x)$. Eliminating x under $x \rightarrow 1$ gives $\theta = J_M / \ln(J_M / S_M)$ to leading order.

Exercise 4.2

Verify that the species–area exponent $z = 0.25$ predicted by neutral theory in $d = 2$ is consistent with a random dispersal kernel of Gaussian form. What changes if the kernel is instead long-tailed (fat-tail exponent μ)?

Solution

With a Gaussian dispersal kernel of variance ℓ^2 , the effective community size in an area A is A/ℓ^2 , and the neutral scaling of the SAD gives $S \sim (A/\ell^2)^{1/4}$. For a power-law kernel $K(r) \sim r^{-1-\mu}$ with $\mu < 2$ the diffusion becomes anomalous and the exponent generalises to $z = \mu / (2\mu + 4)$, matching observation for oceanic communities where $\mu \approx 1$.

4.9. Callouts

i Note

Neutral theory is a *null model*. When it fits, we cannot conclude that species are ecologically equivalent — only that whatever differences exist are not visible in the SAD.

💡 Tip

When analysing a new biodiversity dataset, always fit the neutral model first. A model of niches, competition, or environmental filtering is only worth building if it out-performs the two-parameter neutral baseline.

⚠ Warning

Neutral theory correctly predicts SADs but *not* the correlations between species. When a study reports strong species-species correlations, non-neutral mechanisms are usually needed.

! Important

The scaling exponent z is *robust* — it takes similar values across datasets — but the fitted θ and m are *not*: they depend on the sampling scheme. Report both when publishing.

4.10. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 3. Original neutral-theory reference: Hubbell, *The Unified Neutral Theory of Biodiversity and Biogeography*, Princeton, 2001. Volkov, Banavar, Hubbell & Maritan, *Neutral theory and relative species abundance in ecology*, *Nature* **424**, 1035 (2003).

5. Species interactions and random matrices

Neutral theory (Chapter 3) tells us how much of an ecological pattern can be explained by demographic noise alone. To move beyond it we now add the biological content that neutral theory deliberately ignores: species *interact*. This chapter builds up from the classical Lotka–Volterra system, generalises it to arbitrarily many species, and shows how the celebrated *May stability paradox* leads directly to random matrix theory.

5.1. The Lotka–Volterra equations

For two species — predator y and prey x — the deterministic Lotka–Volterra system is

$$\dot{x} = \alpha x - \beta xy, \quad \dot{y} = \delta xy - \gamma y,$$

with all four rate constants positive. Trajectories are closed orbits around the neutral fixed point $(x^*, y^*) = (\gamma/\delta, \alpha/\beta)$.

Definition 5.1 (Generalised Lotka–Volterra (GLV)). For S species with abundances $\mathbf{x} = (x_1, \dots, x_S)$ the **generalised Lotka–Volterra** equations are

$$\dot{x}_i = x_i \left(r_i + \sum_{j=1}^S \alpha_{ij} x_j \right), \quad i = 1, \dots, S,$$

where r_i is the intrinsic growth rate of species i and α_{ij} is the per-capita effect of species j on species i . Diagonal entries $\alpha_{ii} < 0$ encode intra-specific competition.

Proposition 5.1 (Fixed points of GLV). *The interior fixed point $\mathbf{x}^*$ of GLV satisfies $\mathbf{r} + A\mathbf{x}^* = 0$, i.e. $\mathbf{x}^* = -A^{-1}\mathbf{r}$. Linearising around $\mathbf{x}^*$ gives the Jacobian $J = \text{diag}(\mathbf{x}^*) A$, so stability of $\mathbf{x}^*$ is governed by the eigenvalues of a weighted version of the interaction matrix.*

Remark. GLV is the workhorse model of community ecology because it reproduces the three qualitatively distinct dynamical regimes (fixed point, limit cycle, chaos) with a single interaction matrix. It also emerges as the quasi-steady-state limit of many consumer–resource models.

5.2. Stochastic Volterra model

Adding demographic noise to Lotka–Volterra gives the stochastic model

$$dx_i = x_i \left(r_i + \sum_j \alpha_{ij} x_j \right) dt + \sqrt{2D_i x_i} dW_i(t),$$

whose power spectrum shows *stochastic oscillations* even in parameter regimes where the deterministic model has a stable fixed point.

Theorem 5.2 (Power spectrum of stochastic Lotka–Volterra). *Around a stable interior fixed point $\mathbf{x}^*$, linearised fluctuations $\delta\mathbf{x}(t) = \mathbf{x}(t) - \mathbf{x}^*$ satisfy*

$$d\delta\mathbf{x} = J\delta\mathbf{x}dt + \Sigma^{1/2}d\mathbf{W}(t),$$

with $J = \text{diag}(\mathbf{x}^)A$. The Fourier power spectrum of δx_i is*

$$S_{ii}(\omega) = \sum_{jk} [(J - i\omega I)^{-1}]_{ij} [(J + i\omega I)^{-1}]_{ik} \Sigma_{jk},$$

which typically has a peak at a non-zero ω — the stochastic amplification of a mode that would decay deterministically.

5.3. Random interactions and the May paradox

If $S \gg 1$ the biologist is confronted with S^2 interaction coefficients, which cannot possibly be estimated one by one. Robert May’s insight (1972) was to sidestep the details and ask: *what does a “typical” interaction matrix imply for stability?*

Definition 5.2 (May random-matrix ensemble). Consider a community of S species with random pairwise interactions:

$$A_{ij} = -\delta_{ij} + \sigma c g_{ij}, \quad g_{ij} \sim \mathcal{N}(0, 1) \text{ i.i.d. off-diagonal,}$$

where $c \in [0, 1]$ is the *connectance* (fraction of non-zero entries) and σ the interaction strength. The diagonal encodes self-regulation.

Theorem 5.3 (May’s stability criterion). *The community defined above is stable with high probability in the limit $S \rightarrow \infty$ if and only if*

$$\sigma\sqrt{cS} < 1.$$

Equivalently, if the “complexity” $\sigma\sqrt{cS}$ exceeds one, the interior fixed point loses stability and the community can undergo cycles or extinctions.

The theorem is proved by observing that the eigenvalues of A concentrate on a disc in the complex plane of radius $\sigma\sqrt{cS}$ centred at -1 (the *circular law*), so the rightmost eigenvalue crosses zero exactly at the threshold above.

Corollary 5.4 (Complexity–stability trade-off). *Among the three parameters (S, c, σ) , stability is preserved by decreasing interaction strength or connectance, but not by adding more species. Real ecosystems appear to escape this trade-off by having structured, non-random interaction networks.*

5.4. Proof idea: the circular law

The key tool behind May’s result is the **circular law** of random matrix theory:

Theorem 5.5 (Circular law (Ginibre 1965, Tao–Vu 2010)). *Let M be an $S \times S$ matrix with entries i.i.d. with zero mean and variance $1/S$. In the limit $S \rightarrow \infty$, the empirical spectral distribution of M converges (weakly, almost surely) to the uniform distribution on the unit disc in $\mathbb{C}$.*

Applied to the shifted matrix $A + I$, the circular law gives eigenvalues uniformly on the disc of radius $\sigma\sqrt{cS}$, whose real part is bounded above by that radius.

Lemma 5.6 (Gaussian version). *For a Gaussian M the spectral density can be derived explicitly using a Coulomb-gas / replica calculation. The joint density of eigenvalues in the complex plane is*

$$P(\lambda_1, \dots, \lambda_S) \propto \prod_{i < j} |\lambda_i - \lambda_j|^2 \prod_k e^{-S|\lambda_k|^2}.$$

Interpreted as a 2D Coulomb gas at temperature $1/S$, the distribution flattens into the uniform disc as $S \rightarrow \infty$.

Example 5.1 (Replica trick)

The replica trick evaluates $\langle \ln \det(zI - M) \rangle$ by writing $\ln x = \lim_{n \rightarrow 0} (x^n - 1)/n$, promoting the average of the n -th power of the determinant to a Gaussian integral, and solving the saddle-point at large S . The trick was invented for spin glasses (Sherrington–Kirkpatrick) but reappears everywhere random matrices show up — including in the neural chapters below.

5.5. Structured interactions

Real ecosystems are neither i.i.d. nor mean-field. Two forms of structure matter most:

Proposition 5.7 (Symmetry and correlation). *Let the off-diagonal entries of A be jointly Gaussian with variance σ^2 and correlation $\rho = \text{corr}(A_{ij}, A_{ji})$. Then the spectrum lies on an ellipse of semi-axes $\sigma\sqrt{cS}(1+\rho)$ and $\sigma\sqrt{cS}(1-\rho)$. Predator–prey communities ($\rho < 0$) tend to be more stable than mutualistic ones ($\rho > 0$) for the same complexity.*

Empirical food webs show strongly negative ρ : whenever species j affects i , the reverse effect A_{ji} has the opposite sign. This is one of the main reasons real ecosystems can be much larger than May’s threshold suggests.

5.6. Mutualism, competition, and food webs

Beyond pairwise correlations, network structure — who interacts with whom — is decisive.

Example 5.2 (Nestedness in mutualistic networks)

Empirical plant–pollinator networks are highly *nested*: specialists interact with a subset of the partners of generalists. Nested networks concentrate spectral mass and give a lower principal eigenvalue than random networks of the same S and c , so they can be *more* stable than uncorrelated matrices. Bastolla et al. (2009) argue this explains why mutualistic communities persist at complexities where May’s threshold would predict collapse.

Remark (Adaptive stability). Species evolve. If the interaction matrix A is itself a slow variable driven by selection for stability, the community can *self-organise* into a stable configuration even if a random A of the same statistics would not be. This is the *adaptive* or *evolutionary* form of the complexity–stability question.

5.7. Competitive exclusion

Finally, the classic result of Gause: two species competing for the same limiting resource cannot coexist at equilibrium. In GLV form, if two species i, j have identical growth requirements ($r_i/\alpha_{ii} = r_j/\alpha_{jj}$ and identical impact on the resource), the equilibrium is a one-parameter family, and any small perturbation drives one to extinction. The generalisation to S species and R resources gives:

Theorem 5.8 (Competitive exclusion principle). *In a purely resource-mediated community, the number of coexisting species at equilibrium cannot exceed the number of independent limiting resources: $S_{\text{coex}} \leq R$.*

Nature routinely exceeds this bound (the plankton paradox); the tools of the previous section — noise, non-equilibrium dynamics, network structure — are how the paradox is resolved.

5.8. Exercises

Exercise 5.1

Show that the two-species Lotka–Volterra system is Hamiltonian: find conserved quantities $H(x, y)$ such that $\dot{H} = 0$. Sketch the phase portrait.

Solution

Multiply $\dot{x}/x = \alpha - \beta y$ by δ and $\dot{y}/y = -\gamma + \delta x$ by β and add:

$$\delta \dot{x}/x + \beta \dot{y}/y = \delta \alpha - \gamma \beta,$$

so $H(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y$ is conserved along trajectories. Level sets are closed curves around the fixed point $(x^*, y^*) = (\gamma/\delta, \alpha/\beta)$.

Exercise 5.2

Verify May's threshold numerically: for $S = 100$, $c = 0.5$, sweep σ from 0.05 to 0.3 and estimate the fraction of 10^3 random matrices whose principal eigenvalue crosses zero. Compare with the analytical threshold.

Solution

The theoretical threshold is $\sigma^* = 1/\sqrt{cS} \approx 0.141$. A numerical experiment sampling σg_{ij} with $g_{ij} \sim \mathcal{N}(0, 1)$ shows a sharp transition of the “stable” fraction from ≈ 1 to ≈ 0 within a window $\sigma^* \pm 0.02$, in agreement with the finite- S correction predicted by the Tracy–Widom distribution.

5.9. Callouts

i Note

May's paradox is *not* a proof that ecosystems collapse; it is a proof that *random* ecosystems collapse. The next generation of ecological modelling is about which network structures escape it.

 Tip

When simulating GLV, keep the diagonal of A strictly negative and of order -1 : this fixes the noise scale so that different (S, c, σ) triples are directly comparable.

 Warning

“Complexity” in ecology has three technical meanings that are often conflated: species richness S , connectance c , and interaction strength σ . May’s result is about their product $\sigma\sqrt{cS}$, not about any of them alone.

 Important

The circular law relies on i.i.d. entries. Any structured pattern (block-diagonal, nested, phylogenetically constrained) breaks the assumption and changes the effective threshold. Don’t apply May’s formula to real food-web data without accounting for structure.

5.10. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 4. Original references: May, *Will a large complex system be stable?*, *Nature* **238**, 413 (1972); Allesina & Tang, *Stability criteria for complex ecosystems*, *Nature* **483**, 205 (2012). Random-matrix background: Bordenave & Chafaï, *Around the circular law*, *Probab. Surveys* 9, 1 (2012).

Part II.

Part II: Neural systems

6. Introduction to neural systems

Part I of the course used ecology as the arena for statistical-physics modelling of complex systems. Part II changes scale: from populations of organisms to populations of neurons. This chapter mirrors Prof. Suweis's Chapter 5 and sets the stage.

6.1. 5.1 The brain as a complex system

The brain contains roughly 10^{11} neurons, each with roughly 10^3 synaptic connections — about 10^{14} links in total. No description at the level of an isolated neuron can explain what the brain *does*. Function is emergent: it arises from the interactions.

Definition 6.1 (Signature of complexity in the brain). The brain is a **complex system** in the technical sense of Chapter 1: many simple units, sparse but strong local interactions, ubiquitous noise, and observables (perception, memory, decision) that only exist at the ensemble level.

Two paradigmatic emergent phenomena will occupy the second half of the course.

Example 6.1 (*Neural avalanches*)

Data can be subdivided into time windows $[t_i, t_i + \Delta t]$. An **avalanche** is a burst of activity spanning consecutive windows each with at least one spike. Its **size** s is the number of active neurons summed over the burst. The distribution of s measured in cortical cultures follows a power law

$$p(s) \propto s^{-3/2}$$

over several decades, with a finite-size cutoff at $s \sim N$ (the total number of neurons). We will see in Chapter 10 that this exponent is the mean-field directed-percolation value — evidence for **brain criticality**.

Example 6.2 (*Brain waves*)

Sustained oscillations at specific frequencies — delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), gamma (30–100 Hz) — correlate with distinct cognitive states (sleep, learning, alertness). Sustained oscillations require *synchronisation* of large populations of neurons and are a paradigmatic example of self-organising collective behaviour.

6.2. 5.2 Measurements

Measurements of neural activity span an enormous range of spatial and temporal resolution. It is useful to categorise the tools by their scale.

6.2.1. Non-invasive techniques (whole-brain, coarse)

- **Electroencephalography (EEG).** Electrodes on the scalp measure the local electrical fields produced by large populations of synchronously-firing neurons. Millisecond time resolution but only centimetre spatial resolution; source localisation (the *inverse problem*) is intrinsically ill-posed.
- **Diffusion Tensor Imaging (DTI).** MRI-based technique that tracks the anisotropic diffusion of water along white-matter axon bundles, yielding a mesoscopic map of the anatomical (**structural**) brain connectivity.
- **Functional MRI (fMRI).** Measures the blood-oxygen-level-dependent (BOLD) signal — a proxy for local metabolic demand — with $\sim$ mm spatial resolution but only $\sim$ s temporal resolution. Used to construct **functional connectivity** matrices.

6.2.2. Invasive techniques (few-neuron, fine)

- **Neuro-probes.** Micro-electrodes are inserted into brain tissue and record intracellular or extracellular voltage. Single-cell precision but limited to a few units at a time.
- **In-vitro cultures.** Dissociated neurons are plated on a multi-electrode array; the microscopic dynamics of a small network is observed under controlled conditions.
- **Fluorescence calcium imaging.** Genetically encoded calcium indicators (e.g. GCaMP) fluoresce when a neuron spikes. Modern implementations image thousands of neurons simultaneously at cellular resolution, driving the current revolution in systems neuroscience.

Remark (Multi-scale modelling). The multi-scale measurements motivate multi-scale modelling: Chapter 6 treats a single neuron as an electrical circuit; Chapter 7 treats a *population* of neurons as a rate variable; Chapter 8 introduces plasticity that lets networks learn; Chapter 9 introduces stochastic recurrent networks (Boltzmann machines); Chapter 10 puts everything together and asks whether the healthy brain sits at a critical point.

6.3. Road map for Part II

The rest of the course threads the following logic:

- **Chapter 6 — dynamics of a single neuron.** Structure (soma, dendrites, axon, synapses), membrane potential, ionic pumps, the action potential, spike-count/firing rate, response functions, tuning curves; then bio-electrophysiological models (Hodgkin–Huxley, integrate-and-fire) and the McCulloch–Pitts abstraction.
- **Chapter 7 — firing-rate networks.** Wilson–Cowan and its bifurcations; recurrent networks; linear rate networks and their spectrum; stochastic rate models and avalanche statistics.
- **Chapter 8 — plasticity and Hopfield.** Hebbian rule, subtractive normalisation, associative memory, the Hopfield energy landscape, capacity, and reservoir computing.
- **Chapter 9 — Boltzmann machines.** Stochastic Hopfield dynamics, contrastive divergence, restricted Boltzmann machines.
- **Chapter 10 — brain criticality.** Neural avalanches revisited, the Haldeman–Beggs (HTC) model on structural connectomes, continuous-time approximation, thermodynamic limit.

i Note

The chapter is *phenomenological* — it introduces vocabulary rather than mathematical machinery. All formal statements are proved in the following chapters.

 Tip

When reading the neural literature, always check *which* signal is being measured (spikes vs LFP vs BOLD) and *which* scale is being modelled (single neuron vs population vs whole brain). Cross-scale claims are the source of most controversies.

 Warning

The observation of power-law avalanches, $p(s) \sim s^{-3/2}$, is *necessary but not sufficient* evidence of criticality. Chapter 10 discusses statistical tests (Clauset–Shalizi–Newman) that distinguish true criticality from apparent power laws.

6.4. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 5. On measurement techniques: Dayan & Abbott, *Theoretical Neuroscience*, MIT Press (2005), Chapters 1 and 8. On avalanches: Beggs & Plenz, *Neuronal avalanches in neocortical circuits*, J. Neurosci. **23**, 11167 (2003).

7. Dynamics of a single neuron

Before we build networks we need to understand what one node does. This chapter mirrors Prof. Suweis's Chapter 6: cellular structure and membrane potential; firing rate as a response to a stimulus; the bio-electrophysiological (Hodgkin–Huxley) description; and the McCulloch–Pitts abstraction that will underlie all networks in Chapters 7–9.

7.1. 6.1 The neuron

Neurons are cells specialised for propagating signals over long distances. Their morphology has three functional parts:

- **Dendrites** — highly branched extensions that receive input signals from $\sim 10^4$ other neurons through **synapses**.
- **Soma** (cell body) — houses the nucleus; integrates the incoming currents.
- **Axon** — a single, often very long ($\gtrsim 1$ m in some sensory neurons), extension that carries the output signal to downstream neurons.

Definition 7.1 (Membrane potential). The **membrane potential** V is the electrical potential inside the cell relative to the extracellular medium. Under resting conditions $V_{\text{rest}} \approx -70$ mV. The cell is then said to be **polarised**.

The resting potential is maintained by **ion pumps** and **channels** in the cell membrane that create concentration gradients of Na^+ , K^+ , Ca^{2+} and Cl^- . In particular, K^+ is more concentrated inside the neuron and Na^+ more concentrated outside.

Definition 7.2 (Depolarisation and hyperpolarisation).

- Current in the form of positive ions flowing *out* of the cell (or negative ions flowing *in*) drives V more negative: **hyperpolarisation**.
- Current *into* the cell drives V less negative or positive: **depolarisation**.

When depolarisation pushes V above a threshold $V_{th} \approx -55$ mV, the neuron fires an **action potential** — a stereotyped ~ 100 mV pulse of ~ 1 ms duration that propagates along the axon.

Remark. The action potential's shape is *very nearly the same* every time; the information the neuron transmits is coded in the *timing* of spikes (rate code, temporal code), not in their amplitude. This is the “*all-or-none*” principle established by Adrian in the 1920s and mathematised by Hodgkin & Huxley in the 1950s.

7.2. 6.2 Neuron firing as response to a stimulus

Given an input stimulus $s(t)$ we ask: what is the spike output? Model the spike train as a sum of Dirac deltas:

Definition 7.3 (Neural response function). Given spike times $\{t_i\}_{i=1}^n$ in an interval $[0, T]$, the **neural response function** is

$$\rho(t) = \sum_{i=1}^n \delta(t - t_i).$$

For any test function $h(\tau)$ the convolution

$$\sum_{i=1}^n h(t - t_i) = \int_{-\infty}^{+\infty} h(\tau) \rho(t - \tau) d\tau$$

converts the spike train into a smooth signal.

Definition 7.4 (Spike-count rate and firing rate). The **spike-count rate** on a window $[0, T]$ is

$$r = \frac{n}{T} = \frac{1}{T} \int_0^T \rho(\tau) d\tau.$$

The **time-dependent firing rate** $\phi(t)$ is the running average of ρ over a moving window; formally

$$\phi(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \mathbb{E} \left[\int_{t-\Delta t/2}^{t+\Delta t/2} \rho(\tau) d\tau \right].$$

For a stationary point process $\phi(t) = \lambda$ is the intensity.

Example 7.1 (*Tuning curve of a sensory neuron*)

For a sensory neuron responding to an oriented visual stimulus, the firing rate as a function of orientation θ is a bell-shaped **tuning curve**:

$$\phi(\theta) = \phi_{\max} \exp(-(\theta - \theta^*)^2 / (2\sigma^2)),$$

where θ^* is the preferred orientation. Cataloguing tuning curves is a workhorse method of systems neuroscience.

7.3. 6.3 Bio-electrophysiological modelling of the neuron

To go beyond phenomenology we need a physical model of the membrane. The membrane behaves as a capacitor with parallel-conductance ion channels:

Definition 7.5 (RC equivalent circuit). For a passive patch of membrane,

$$C_m \frac{dV}{dt} = -g_L(V - V_L) + I(t),$$

where C_m is the membrane capacitance, g_L the leak conductance, V_L the leak reversal potential, and $I(t)$ the input current. In the absence of active channels, the membrane behaves as a low-pass filter with time constant $\tau_m = C_m/g_L \approx 10\text{--}30$ ms.

Theorem 7.1 (Hodgkin–Huxley model). *Adding voltage-gated Na^+ and K^+ conductances, whose gating*

variables m, h, n obey first-order kinetics, gives

$$C_m \frac{dV}{dt} = -g_{Na} m^3 h (V - V_{Na}) - g_K n^4 (V - V_K) - g_L (V - V_L) + I(t),$$

with $\dot{m} = \alpha_m(V)(1 - m) - \beta_m(V)m$, and analogous for h, n . The four-dimensional system exhibits an excitable regime with an action potential and a stable refractory period.

The Hodgkin–Huxley model won its authors the 1963 Nobel Prize in Physiology and remains the biophysical gold standard.

Proposition 7.2 (Leaky integrate-and-fire). *For network-scale modelling the Hodgkin–Huxley description is impractical. The **leaky integrate-and-fire (LIF)** neuron replaces the spike-generating dynamics by a threshold: integrate the passive RC equation until V crosses V_{th} , emit a spike, reset $V \rightarrow V_{reset}$, and impose a refractory period τ_{ref} . LIF captures the essential firing behaviour with a single ODE and one non-linearity.*

Corollary 7.3 (LIF firing rate under constant current). *For a LIF neuron driven by a constant current $I_0 > (V_{th} - V_L)g_L$, the interspike interval is*

$$\Delta t = \tau_m \ln \left(\frac{I_0/g_L - (V_{reset} - V_L)}{I_0/g_L - (V_{th} - V_L)} \right),$$

so the firing rate is $\phi(I_0) = 1/\Delta t$. This is the rate transfer function used in Wilson–Cowan (Chapter 7).

7.4. 6.4 McCulloch–Pitts artificial neuron

For network-scale mathematical modelling we abstract even further. McCulloch & Pitts (1943) proposed the following:

Definition 7.6 (McCulloch–Pitts neuron). A **McCulloch–Pitts neuron** with N inputs $x_1, \dots, x_N$ (each 0 or 1), synaptic weights $w_1, \dots, w_N$, and threshold θ produces output

$$y = \Theta \left(\sum_{i=1}^N w_i x_i - \theta \right) \in \{0, 1\},$$

where Θ is the Heaviside step function.

The McCulloch–Pitts unit is the fundamental building block of every neural network in this course.

Example 7.2 (Logic gates from McCulloch–Pitts)

With appropriate weights and threshold, a single McCulloch–Pitts unit computes any two-input Boolean function *except* XOR. A **two-layer** network of such units can compute XOR — the classic Minsky–Papert observation that motivated multi-layer networks.

Remark. The step-function activation of the McCulloch–Pitts unit was replaced in modern practice by the sigmoid $\sigma(x) = 1/(1 + e^{-x})$ (allowing gradient-based learning) and later by ReLU $\max(0, x)$ (avoiding vanishing gradients in deep networks). Both are approximations of the biological non-linearity.

7.5. Exercises

Exercise 7.1

Derive the LIF firing rate under constant current given above. State the parameter regime where the neuron fires periodically.

Solution

Solve $C_m \dot{V} = -g_L(V - V_L) + I_0$ with initial condition $V(0) = V_{reset}$ and first-passage condition $V(\Delta t) = V_{th}$. The solution is $V(t) = V_\infty + (V_{reset} - V_\infty)e^{-t/\tau_m}$ with $V_\infty = V_L + I_0/g_L$. Setting $V(\Delta t) = V_{th}$ and solving for Δt gives the formula. The neuron fires periodically iff $V_\infty > V_{th}$, i.e. $I_0 > (V_{th} - V_L)g_L$.

Exercise 7.2

Show that no single McCulloch–Pitts unit implements the XOR function on two inputs.

Solution

Suppose weights w_1, w_2 and threshold θ give XOR. Then $w_1 + w_2 \leq \theta - 1$ (input (1, 1) outputs 0), but each of $w_1, w_2 > \theta - 1$ (inputs (1, 0) and (0, 1) output 1). Summing: $w_1 + w_2 > 2\theta - 2$. Combining with $w_1 + w_2 \leq \theta - 1$: $\theta - 1 > 2\theta - 2$, i.e. $\theta < 1$. But also (0, 0) must output 0, so $\theta > 0$. Both bounds together give $0 < \theta < 1$. Now check (0, 1): need $w_2 > \theta > 0$, and (1, 1): $w_1 + w_2 \leq \theta - 1 < 0$. So $w_1 < -w_2 < -\theta < 0$. Then (1, 0): $w_1 < 0 < \theta$, so output is 0, not 1. Contradiction.

Exercise 7.3

Verify that the leak time constant $\tau_m = C_m/g_L$ has units of time given typical values $C_m \sim 10^{-6} \text{ F/cm}^2$ and $g_L \sim 10^{-4} \text{ S/cm}^2$.

Solution

$C/g = \text{F/S} = (\text{C/V})/(1/\Omega) = \text{C} \cdot \Omega/\text{V} = \text{s}$ (since $\text{V} = \Omega \text{ A} = \Omega \text{ C/s}$). Numerically $\tau_m = 10^{-2} \text{ s} = 10 \text{ ms}$, matching biology.

7.6. Callouts

Note

The three levels of description — Hodgkin–Huxley, integrate-and-fire, McCulloch–Pitts — represent decreasing biological fidelity but increasing analytical tractability. Which one to use depends on the question you are asking.

Tip

For any rate model of a network, first sketch the transfer function $\phi(I)$ of a single neuron. Its shape (linear-rectifier, sigmoidal, threshold-linear) will determine the qualitative regimes of the network.

! Important

The action potential is *stereotyped*: its amplitude carries no information. This is the reason firing-rate models exist — throwing away the details of each spike loses little.

⚠ Warning

The LIF neuron has a hard reset that removes any memory of the spike time; a soft-reset LIF (subtract $V_{th} - V_{reset}$) can produce very different collective dynamics near the firing threshold.

7.7. Book supplements

The bio-electrophysiological content of this chapter is treated in canonical detail in Dayan & Abbott, *Theoretical Neuroscience*, MIT Press (2005), Chapters 5 and 6. The original references are Hodgkin & Huxley, *A quantitative description of membrane current and its application to conduction and excitation in nerve*, J. Physiol. **117**, 500 (1952); McCulloch & Pitts, *A logical calculus of the ideas immanent in nervous activity*, Bull. Math. Biophys. **5**, 115 (1943).

7.8. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 6. Dayan & Abbott, Chapters 5–6. Original references cited above.

8. Firing-rate networks

We now switch scale: from populations of species to populations of neurons. The ecological toolkit — mean-field ODEs, master equations, mean-field random matrices — carries over almost unchanged, and the first stop is the *firing-rate model*, the simplest description of neural population activity.

8.1. Rate versus spike models

Individual neurons emit action potentials — stereotyped voltage pulses of ~ 1 ms duration. When many neurons fire the informative signal is not the timing of each spike but the *rate* $\phi(t)$ at which spikes are emitted in a short window. Firing-rate models replace the discrete spike train by this continuous rate.

Definition 8.1 (Firing rate). The **firing rate** of a neuron is

$$\phi(t) = \frac{\text{number of spikes in a window } [t - \Delta/2, t + \Delta/2]}{\Delta},$$

in the limit $\Delta \rightarrow 0$ after ensemble-averaging over trials. For a stationary point process $\phi(t) = \lambda$ is the intensity of the spike train.

Remark. For most biophysical purposes the rate description is valid whenever the network activity varies on timescales $\gg 1/\lambda$. Below that scale (single-spike coincidence coding) rate models fail and one must go back to spike-timing models such as leaky integrate-and-fire.

8.2. Excitatory–inhibitory populations

Even before we introduce networks, the effective single-population equation is instructive. Let $\phi(t)$ be the mean rate of a homogeneous population; the deterministic *rate equation* is

$$\tau \dot{\phi} = -\phi + F(I),$$

where I is the input current and F is a monotone gain function (the *transfer function*), typically sigmoidal.

The next level of realism separates the population into *excitatory* (E) and *inhibitory* (I) sub-populations that interact via signed synapses:

Definition 8.2 (Wilson–Cowan equations). The **Wilson–Cowan equations** for the mean firing rates of coupled excitatory and inhibitory populations are

$$\begin{aligned}\tau_E \dot{\phi}_E &= -\phi_E + F_E(W_{EE}\phi_E - W_{EI}\phi_I + I_E^{ext}), \\ \tau_I \dot{\phi}_I &= -\phi_I + F_I(W_{IE}\phi_E - W_{II}\phi_I + I_I^{ext}).\end{aligned}$$

The four constants $W_{XY} > 0$ encode the average synaptic weights between populations; $F_{E,I}$ are sigmoid transfer functions saturating at some maximum rate.

Theorem 8.1 (Fixed points and oscillations of Wilson–Cowan). *The Wilson–Cowan system exhibits three qualitatively different regimes: (i) a single stable low-activity fixed point (rest state); (ii) two stable fixed points separated by a saddle (bistability, working memory); (iii) a Hopf-bifurcated limit cycle (oscillation). The transitions occur along a cusp in the (I_E, I_I) external-input plane.*

Example 8.1 (*Gamma oscillations*)

Wilson–Cowan networks with excitatory time constant $\tau_E \sim 5$ ms and inhibitory $\tau_I \sim 10$ ms show a Hopf bifurcation to oscillations at ~ 40 Hz — the *gamma rhythm* implicated in perceptual binding and attention. The oscillation frequency depends on the *lag* between E and I responses, not on any individual neuron’s intrinsic frequency.

8.3. Recurrent networks

A more microscopic description replaces the two lumped populations by N individual rate units:

$$\tau \dot{\phi}_i = -\phi_i + F\left(\sum_{j=1}^N W_{ij}\phi_j + I_i^{ext}\right), \quad i = 1, \dots, N.$$

When W is a full matrix this is a *recurrent* rate network — the object of the rest of the chapter.

8.4. Linear recurrent networks

Suppose F is linear, $F(x) = x$. Then the dynamics is

$$\tau \dot{\phi} = (W - I)\phi + \mathbf{I}^{ext}.$$

The behaviour is governed by the spectrum of W :

Proposition 8.2 (Stability of linear recurrent nets). *The linear rate network is stable if and only if the leading eigenvalue of W has real part less than 1. If W is drawn from the May ensemble of Chapter 4 with parameters (N, c, σ) , the leading eigenvalue is $\sigma\sqrt{cN}$; the network is stable iff $\sigma\sqrt{cN} < 1$, exactly the ecological threshold.*

The equivalence between May’s stability criterion and the transition to *chaos* in random recurrent networks is one of the most striking cross-domain results in the notes: Sompolsky, Crisanti and Sommers (1988) proved that a nonlinear random rate network transitions from a fixed point to deterministic chaos exactly at $\sigma\sqrt{N} = 1$.

Theorem 8.3 (Sompolsky–Crisanti–Sommers transition). *For a nonlinear rate network with W_{ij} i.i.d. Gaussian of variance σ^2/N and gain function $\phi = \tanh(x)$, the fixed point at zero loses stability at $\sigma = 1$. For $\sigma > 1$ the dynamics is deterministically chaotic, with a positive Lyapunov exponent that grows continuously from zero.*

8.5. Stochastic rate networks

Adding noise to the rate equation gives a Langevin picture:

$$d\phi_i = \tau^{-1}(-\phi_i + F(\sum_j W_{ij}\phi_j)) dt + \sigma_\eta dW_i(t).$$

For a linear F the process is Ornstein–Uhlenbeck, and its power spectrum is

$$S(\omega) = \frac{2\sigma_\eta^2 \tau^2}{|(W - I)\tau + i\omega\tau I|^2 + \omega^2 \tau^2},$$

generalising the stochastic-Volterra result of Chapter 4.

Lemma 8.4 (Avalanches from stochastic linear rate models). *Near the stability boundary $\sigma\sqrt{cN} = 1$, the linear stochastic rate network shows avalanches of bursty activity whose size distribution follows a power law $P(s) \sim s^{-3/2}$, characteristic of a critical branching process. This has been proposed as a mechanism for the empirically observed avalanche statistics in cortical recordings.*

8.6. Exercises

Exercise 8.1

Derive the mean-field fixed point of a Wilson–Cowan network with symmetric parameters $\tau_E = \tau_I$, $W_{EE} = W_{II} = 0$, $W_{EI} = W_{IE} = w > 0$, and no external input. When does a Hopf bifurcation occur as w is increased?

Solution

The fixed points satisfy $\phi_E = F(-w\phi_I)$ and $\phi_I = F(w\phi_E)$. If F is odd, $\phi_E = -\phi_I$ is a solution which, for $F(x) = \tanh(x)$, gives $\phi_E = \tanh(w\phi_E)$: only $\phi_E = 0$ if $w < 1$; two symmetric nontrivial roots for $w > 1$. Linearising around $(0, 0)$: eigenvalues of the Jacobian are $\pm iwF'(0)$, so at $wF'(0) = 1$ we have a *pure imaginary* pair — the Hopf point. For $w > w_c$ a limit cycle grows out of it.

Exercise 8.2

Show that the transition to chaos in the Sompolinsky–Crisanti–Sommers network can be understood as a growth of the largest Lyapunov exponent from zero. Use the dynamic mean-field theory to write a self-consistent equation for the autocorrelation $C(t) = \langle \phi_i(t')\phi_i(t' + t) \rangle$.

Solution

The dynamic mean-field ansatz replaces the recurrent input to any neuron by a Gaussian field with the same auto-covariance. Writing $\phi_i(t) = F(x_i(t))$ with $\langle x_i(t)x_i(t') \rangle = \sigma^2 C_\phi(t - t')$, self-consistency gives

$$C_\phi(\tau) = \int \tanh(\sqrt{\sigma^2 C_\phi(0)} z_1) \tanh(\sqrt{\sigma^2 C_\phi(0)} z_2) d\mathcal{N},$$

which is a fixed-point equation for the correlation function. Its trivial solution $C_\phi \equiv 0$ becomes unstable at $\sigma = 1$; a nontrivial oscillating solution grows continuously above threshold, matching the Lyapunov diagnosis.

8.7. Callouts

Note

Firing-rate models describe *populations*, not single neurons. Predictions such as gamma oscillations or avalanches are about *ensembles*; single-neuron voltage traces look nothing like their solutions.

Tip

The single most useful check on any recurrent-network simulation is the eigenvalue spectrum of W : if the leading eigenvalue's real part exceeds one, the linear part is already unstable and the dynamics you are seeing is dominated by the nonlinearity, not by the network you intended to model.

Warning

The result “power-law avalanches with exponent $-3/2$ ” appears in many neuroscience datasets. It is *not* by itself evidence of criticality: a wide variety of non-critical processes produce the same exponent.

8.8. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 7. Dayan & Abbott, *Theoretical Neuroscience*, Ch. 7; Wilson & Cowan, *Excitatory and inhibitory interactions in localized populations of model neurons*, Biophys. J. **12**, 1 (1972); Sompolinsky, Crisanti & Sommers, *Chaos in random neural networks*, Phys. Rev. Lett. **61**, 259 (1988).

9. Plasticity, associative memory and Hopfield networks

The synapses W_{ij} of Chapter 7 were fixed. In real brains they *change* — that is *plasticity* — and the pattern of changes lets the network *store and retrieve* information. This chapter mirrors Prof. Suweis’s Chapter 8: Hebbian plasticity and its subtractively-normalised variant, associative-memory networks, and the Hopfield model with its statistical-mechanical analysis. The extension to *stochastic* recurrent networks (Boltzmann machines) is treated separately in Chapter 9.

9.1. Hebbian plasticity

The essence of Hebb’s rule is captured in the slogan “neurons that fire together, wire together”. Formally:

Definition 9.1 (Hebbian plasticity rule). Let ϕ_i, ϕ_j be the firing rates of two neurons and W_{ij} the weight of the synapse from j to i . The **Hebbian rule** is

$$\tau_W \dot{W}_{ij} = \eta \phi_i \phi_j,$$

with learning rate $\eta > 0$. Under repeated correlated activity of pre- and post-synaptic neurons, W_{ij} grows without bound.

Remark. The unbounded growth is unphysical: real synaptic weights are bounded. Two standard fixes are (i) *subtractive normalisation* — subtract the mean activity from each factor; (ii) *multiplicative normalisation* (Oja’s rule) — subtract a term proportional to $\phi_i^2 W_{ij}$ that stabilises the norm.

Proposition 9.1 (Subtractive normalisation and PCA). *Under the subtractively normalised Hebbian rule with a linear rate unit driven by inputs $\mathbf{x}(t)$, the synaptic weight vector converges to the leading eigenvector of the input covariance $\Sigma = \langle \mathbf{x}\mathbf{x}^\top \rangle$. Hebbian plasticity therefore performs principal component analysis.*

9.2. Associative memory networks

Suppose we want the network to *remember* a set of patterns $\{\xi^\mu\}_{\mu=1}^P$, each a vector of N binary units $\xi_i^\mu \in \{-1, +1\}$. The Hebbian prescription is:

Definition 9.2 (Hebbian storage rule). Given patterns ξ^μ , the **Hebbian synaptic matrix** is

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu, \quad W_{ii} = 0.$$

The network dynamics is $s_i(t+1) = \text{sgn}(\sum_j W_{ij} s_j(t))$. A pattern ξ^μ is *stored* if it is a fixed point of these dynamics.

This construction is the **Hopfield network**. Its analysis, unlike the recurrent nets of Chapter 5, is *statistical-mechanical*: the dynamics has a Lyapunov function, so a temperature can be introduced by hand and the equilibrium ensemble studied.

9.3. Hopfield energy landscape

Theorem 9.2 (Lyapunov function of Hopfield dynamics). *For the Hopfield model with symmetric weights $W_{ij} = W_{ji}$ and zero diagonal, the quantity*

$$H[\mathbf{s}] = -\frac{1}{2} \sum_{ij} W_{ij} s_i s_j$$

is monotonically non-increasing under synchronous or asynchronous sign updates. Fixed points are local minima of H .

The stored patterns ξ^μ are, by construction, minima of H . But the Hebbian rule also creates **spurious minima** — combinations like $\text{sgn}(\xi^{\mu_1} + \xi^{\mu_2} + \xi^{\mu_3})$ that are also local minima of H , but do not correspond to any single stored pattern.

Proposition 9.3 (Hopfield capacity). *The maximum number of patterns that can be stored and retrieved reliably in a Hopfield network of N binary units is*

$$P_{\max} \approx 0.138 N.$$

Beyond this critical loading $\alpha_c = P/N \approx 0.138$, retrieval fails: a spin-glass phase takes over in which memory patterns cease to be attractors.

The threshold α_c is not a hand-wavy back-of-the-envelope result; it comes from the *replica-symmetric* solution of the Hopfield model treated as a spin glass.

9.4. Statistical mechanics of the Hopfield model

At temperature $T = 1/\beta$ the Hopfield model is a disordered Ising system with couplings W_{ij} built from random patterns. The partition function is

$$Z(\beta, \{\xi\}) = \sum_{\mathbf{s}} e^{-\beta H[\mathbf{s}]}.$$

Averaging $\ln Z$ over the quenched randomness of the patterns (as in Chapter 4's replica trick) gives the free energy in terms of two order parameters: the *overlaps* $m^\mu = \frac{1}{N} \sum_i \xi_i^\mu s_i$ with each stored pattern, and the *Edwards–Anderson* order parameter $q = \frac{1}{N} \sum_i \langle s_i \rangle^2$ measuring frozen local magnetisation.

Theorem 9.4 (Amit–Gutfreund–Sompolinsky phase diagram). *The (T, α) phase diagram of the Hopfield model has three phases:*

1. **Retrieval phase** at low T and $\alpha < \alpha_c(T)$: one m^μ is $O(1)$, memories are attractors.
2. **Spin-glass phase** at low T and $\alpha > \alpha_c(T)$: all $m^\mu = 0$ but $q > 0$; memories are lost.
3. **Paramagnetic phase** at high T : $q = 0$; no correlations survive.

At $T = 0$ the critical loading is $\alpha_c = 0.138$.

Example 9.1 (*Retrieval as a physical spin system*)

Fixing $\alpha < 0.138$ and cooling from high T , the Hopfield network undergoes a *second-order* transition from paramagnet to retrieval phase at $T_c(\alpha)$. Below T_c the network settles to a state with macroscopic overlap with a randomly chosen stored pattern. This is the physical mechanism of associative memory.

9.5. Supervised learning with recurrent networks (reservoir computing)

An alternative to storing static patterns is to use the *transient* dynamics of a randomly connected recurrent network to represent inputs. This is the idea of *reservoir computing* (also known as an echo-state network or liquid-state machine):

Definition 9.3 (Reservoir computing). Fix a large, random, sub-critical recurrent network with weights W (with $\sigma\sqrt{N} < 1$ so activity decays). Drive it with a time-varying input $\mathbf{u}(t)$. Train *only* a linear readout $\mathbf{y} = C\phi(t)$ to match a target signal $\mathbf{y}^*(t)$.

Because only the readout is trained, the optimisation is convex (least squares) and does not suffer from vanishing gradients. Reservoirs poised near the edge of chaos ($\sigma\sqrt{N} \approx 1$) have the highest computational capacity — an application of the criticality theme from Chapter 4.

9.6. Toward stochastic recurrent networks

The Hopfield model is *deterministic*: given the initial state, the trajectory is uniquely determined by the sign updates. Making it stochastic — the update is drawn from a Boltzmann distribution over the two options — gives the Boltzmann machine and connects the associative-memory picture to modern energy-based generative models. This extension is the subject of Chapter 9.

9.7. Exercises

Exercise 9.1

Verify that Oja's rule $\dot{W} = \eta(\phi\mathbf{x} - \phi^2 W)$ (with $\phi = W^\top \mathbf{x}$) drives W to unit norm and to the principal eigenvector of the input covariance.

Solution

Compute $|\dot{W}|^2 = 2W^\top \dot{W} = 2\eta(\phi W^\top \mathbf{x} - \phi^2 |W|^2) = 2\eta\phi^2(1 - |W|^2)$: the norm relaxes to one. On the sphere, the fixed points satisfy $\Sigma W = \lambda W$ with $\lambda = W^\top \Sigma W$. Stability analysis shows that only the top eigenvector is stable.

Exercise 9.2

Show that the Hopfield capacity $\alpha_c \approx 0.138$ can be derived heuristically by demanding that the noise contribution to the local field on a pattern be smaller than the signal. Give the order-of-magnitude argument.

Solution

The local field on pattern μ at site i is $h_i^\mu = \xi_i^\mu + \frac{1}{N} \sum_{\nu \neq \mu} \sum_j \xi_i^\nu \xi_j^\nu \xi_j^\mu$. The signal is $\xi_i^\mu = \pm 1$ (variance 1); the noise is a sum of $P - 1$ terms each of variance $1/N$, so its variance is $(P - 1)/N \sim \alpha$. Retrieval requires signal-to-noise > 1 , giving $\alpha \lesssim 1$. The precise number 0.138 comes from a more careful replica calculation that accounts for correlations among the noise terms.

9.8. Callouts

i Note

Hebbian plasticity is one of very few learning rules that is *both* biologically plausible and mathematically well-behaved. Most modern deep-learning algorithms (backprop) are neither.

💡 Tip

When implementing a Hopfield network from scratch, always plot the energy $H[\mathbf{s}(t)]$ during dynamics: monotone decrease is a necessary correctness check.

! Important

The Hopfield capacity $\alpha_c \approx 0.138$ is the number of *uncorrelated* patterns that can be stored. Correlated patterns (e.g. real images) have a much lower effective capacity.

⚠ Warning

Real synapses are sign-constrained (Dale's law) and typically bounded above by a saturation weight. The mathematical Hopfield model ignores both. Introducing these constraints reduces capacity and can change the phase diagram — see Brunel & Nadal (1998).

9.9. Book supplements

The statistical-mechanical analysis of Hopfield networks is treated in canonical detail in Hertz, Krogh & Palmer, *Introduction to the Theory of Neural Computation*, Westview (1991), Chapters 2 and 10. For a modern perspective on associative memory and its capacity, Krotov & Hopfield, *Dense associative memory is robust to adversarial inputs*, Neural Comput. **30**, 3151 (2018).

9.10. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 8. Hopfield, *Neural networks and physical systems with emergent collective computational abilities*, PNAS **79**, 2554 (1982); Amit, Gutfreund & Sompolinsky, *Statistical mechanics of neural networks near saturation*, Ann. Phys. **173**, 30 (1987). Hertz, Krogh & Palmer, Chapters 2, 10.

10. Boltzmann machines

Chapter 8 introduced the Hopfield model as a *deterministic* recurrent network with a Lyapunov function. Making its update stochastic gives a full *probabilistic* recurrent model whose stationary distribution is the Gibbs measure. This is the Boltzmann machine — an energy-based generative model that underlies modern probabilistic deep learning. The chapter mirrors Prof. Suweis's Chapter 9.

10.1. 9.1 Stochastic recurrent networks

The deterministic Hopfield update maps a state $\mathbf{s}^{(t)}$ to $s_i^{(t+1)} = \text{sgn}(h_i)$, where $h_i = \sum_j W_{ij}s_j$ is the local field. The **stochastic** update samples the new value from a Boltzmann distribution over the two options:

Definition 10.1 (Glauber (heat-bath) dynamics). At each step, choose a spin i (uniformly at random) and update

$$P(s_i^{(t+1)} = +1) = \frac{1}{1 + e^{-2\beta h_i}}, \quad P(s_i^{(t+1)} = -1) = \frac{1}{1 + e^{+2\beta h_i}},$$

where $\beta = 1/T$ is the inverse temperature and $h_i = \sum_{j \neq i} W_{ij}s_j^{(t)} + b_i$ is the local field including any bias b_i .

Theorem 10.1 (Detailed balance and Gibbs equilibrium). *Glauber dynamics satisfies detailed balance with respect to the Boltzmann distribution*

$$\pi(\mathbf{s}) = \frac{1}{Z} \exp(-\beta H[\mathbf{s}]), \quad H[\mathbf{s}] = -\frac{1}{2} \sum_{ij} W_{ij}s_i s_j - \sum_i b_i s_i.$$

Any ergodic Glauber chain therefore has π as its unique stationary distribution.

At $T \rightarrow 0$ the sigmoid becomes a step function and Glauber dynamics reduces to the deterministic Hopfield update. At $T \rightarrow \infty$ the update is uniformly random and all correlations are lost. Intermediate temperatures give the interesting regime of *stochastic associative memory*.

Remark. Simulated annealing — running Glauber dynamics with a decreasing schedule $T(t) \rightarrow 0$ — is a general-purpose optimisation algorithm for combinatorial problems whose cost function can be written as a Hopfield energy. The algorithm converges to a global minimum in the limit of very slow cooling.

10.2. 9.2 The Boltzmann machine

Making the recurrent stochastic network *trainable* means adjusting weights and biases so that the stationary distribution matches a target data distribution.

Definition 10.2 (Boltzmann machine). A **Boltzmann machine** is a network of binary units $s_i \in \{0, 1\}$

with joint Gibbs distribution

$$P(\mathbf{s}; W, \mathbf{b}) = \frac{1}{Z(W, \mathbf{b})} \exp\left(\sum_{i < j} W_{ij} s_i s_j + \sum_i b_i s_i\right).$$

Units are split into **visible** units $\mathbf{v}$ (observed) and **hidden** units $\mathbf{h}$ (latent). Training means adjusting $(W, \mathbf{b})$ to make the visible marginal $P(\mathbf{v})$ close to the empirical data distribution $\hat{p}(\mathbf{v})$.

The negative log-likelihood gradient with respect to W_{ij} has an elegant form.

Theorem 10.2 (Boltzmann-machine gradient). *The log-likelihood gradient decomposes into a data-driven and a model-driven term:*

$$\frac{\partial \log P(\mathbf{v})}{\partial W_{ij}} = \langle s_i s_j \rangle_{\text{data}} - \langle s_i s_j \rangle_{\text{model}}.$$

The first expectation is over the data (with hidden units integrated out under the current model); the second is over the full model distribution.

Remark (Why training is hard). The model-driven expectation requires sampling from the full Boltzmann machine — which in general requires equilibrating a Glauber chain whose mixing time can be exponential in the number of hidden units. This is the historical obstacle to training general Boltzmann machines.

10.3. 9.3 Restricted Boltzmann machines

The intractability of general Boltzmann-machine training led Smolensky (1986) and Hinton (2002) to constrain the graph:

Definition 10.3 (Restricted Boltzmann machine (RBM)). A **restricted Boltzmann machine** has visible units $\mathbf{v} \in \{0, 1\}^{N_v}$ and hidden units $\mathbf{h} \in \{0, 1\}^{N_h}$ with connections *only* between the two layers (bipartite graph). Its energy function is

$$H[\mathbf{v}, \mathbf{h}] = -\mathbf{v}^\top W \mathbf{h} - \mathbf{a}^\top \mathbf{v} - \mathbf{b}^\top \mathbf{h},$$

and the joint distribution is Gibbs with $-\beta H$.

Proposition 10.3 (Conditional independence in an RBM). *In an RBM, all visible units are conditionally independent given the hidden units (and vice versa):*

$$P(\mathbf{v} \mid \mathbf{h}) = \prod_i \sigma(a_i + \mathbf{W}_i^\top \mathbf{h})^{v_i} (1 - \sigma(\cdot))^{1-v_i},$$

and analogously for $P(\mathbf{h} \mid \mathbf{v})$. Consequently Gibbs sampling can alternate layer-wise: sample all visibles in parallel, then all hidden.

The conditional-independence structure is what makes RBMs tractable.

10.4. 9.4 Contrastive divergence

Even for an RBM, computing the model-driven gradient exactly requires a well-mixed sampler. Hinton's **contrastive divergence** (CD- k) algorithm approximates it by running only k steps of Gibbs sampling

starting from the data.

Definition 10.4 (Contrastive divergence (CD- k)). For each mini-batch of data $\{\mathbf{v}^{(n)}\}$:

1. Compute hidden activations $\mathbf{h}^{(0)}$ by sampling from $P(\mathbf{h} \mid \mathbf{v}^{(n)})$.
2. Iterate k Gibbs steps: $\mathbf{v}^{(t+1)} \sim P(\mathbf{v} \mid \mathbf{h}^{(t)})$, $\mathbf{h}^{(t+1)} \sim P(\mathbf{h} \mid \mathbf{v}^{(t+1)})$.
3. Update weights: $\Delta W_{ij} \propto \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{k\text{-step}}$.

CD-1 (single Gibbs step) is what made RBM training practical and was the algorithm behind the 2006 deep-learning revival.

Example 10.1 (*Stacked RBMs and deep belief networks*)

Training a deep belief network layer-by-layer with RBM pretraining was the first successful method for training deep neural networks (Hinton, Osindero & Teh, 2006). It has been largely superseded by end-to-end backpropagation with modern initialisation schemes, but the *energy-based* framework it introduced underlies modern diffusion and score-based generative models.

10.5. 9.5 Connection to statistical mechanics

The Boltzmann machine is *literally* an Ising-like spin system. Its analysis borrows heavily from the equilibrium statistical mechanics we used in Chapter 8:

- The free energy $F = -T \log Z$ is the object minimised by the model at fixed data.
- The magnetisations $\langle s_i \rangle$ are the model's mean predictions.
- The susceptibility $\chi_{ij} = \partial \langle s_i \rangle / \partial b_j$ is the Fisher information of the network.
- The training gradient is the difference of two magnetisations — a fluctuation–dissipation-like identity.

Remark (Deep learning as statistical mechanics). The vocabulary of energy-based models — free energy, partition function, temperature — is not decorative; it is what makes the training procedure a physically motivated learning algorithm. Modern *score matching* and *denoising diffusion* generative models are direct descendants of the RBM framework.

10.6. Exercises

Exercise 10.1

Show that in the low-temperature limit ($\beta \rightarrow \infty$) the Glauber dynamics of a symmetric Boltzmann machine reduces to the deterministic Hopfield update.

Solution

The sigmoid $\sigma(2\beta h_i) \rightarrow \Theta(h_i)$ as $\beta \rightarrow \infty$: the update chooses $s_i = +1$ with probability 1 if $h_i > 0$ and 0 otherwise. This is exactly the sign rule of the deterministic Hopfield update.

Exercise 10.2

Derive the Boltzmann-machine gradient formula from the log-likelihood $\log P(\mathbf{v}) = \log \sum_{\mathbf{h}} e^{-\beta H(\mathbf{v}, \mathbf{h})} - \log Z$.

Solution

Differentiate: the first term gives $\langle s_i s_j \rangle_{P(\mathbf{h}|\mathbf{v})}$ (data-conditioned expectation), the second gives $\langle s_i s_j \rangle_{P(\mathbf{v}, \mathbf{h})}$ (unconditioned model expectation). Difference is the negative gradient.

Exercise 10.3

Show that in an RBM the marginal $P(\mathbf{v})$ is not itself a Boltzmann distribution over $\mathbf{v}$: the hidden marginalisation induces non-pairwise (higher-order) interactions.

Solution

$P(\mathbf{v}) \propto \sum_{\mathbf{h}} \exp(-H)$. For binary hidden units, $\sum_{h_j \in \{0,1\}} \exp(h_j(b_j + \mathbf{W}_j^\top \mathbf{v})) = 1 + \exp(b_j + \mathbf{W}_j^\top \mathbf{v})$, so $P(\mathbf{v}) \propto \exp(\mathbf{a}^\top \mathbf{v}) \prod_j (1 + e^{b_j + \mathbf{W}_j^\top \mathbf{v}})$. The log of the product term contains $\log(1 + e^{\cdot})$ which is not linear in $\mathbf{v}$; expanding gives all orders of interactions among the v_i .

10.7. Callouts

i Note

Boltzmann machines are the cleanest bridge between statistical mechanics and generative modelling. Every energy-based model in modern ML (score matching, diffusion) is a direct descendant.

💡 Tip

When implementing an RBM, initialise weights small (e.g. $\mathcal{N}(0, 0.01)$) and use a small learning rate. Contrastive divergence is unstable if the Gibbs chain saturates.

! Important

Restricted Boltzmann machines are tractable *only* because of the bipartite graph. Any non-bipartite variant (three-way units, deep BM) loses conditional independence and requires more sophisticated inference.

⚠ Warning

CD-1 is *biased*: the gradient it computes is not exactly the gradient of the log-likelihood. It works well in practice for shallow RBMs but should be replaced by longer Gibbs chains (PCD, CD-25) for deeper models.

10.8. Book supplements

The standard reference on Boltzmann machines and restricted variants is Hinton, *A practical guide to training restricted Boltzmann machines*, in Montavon, Orr & Müller (eds.), *Neural Networks: Tricks of the Trade*, Springer (2012). For the statistical mechanics of learning, MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge (2003), Chapter 43.

10.9. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 9. Ackley, Hinton & Sejnowski, *A learning algorithm for Boltzmann machines*, *Cognitive Science* **9**, 147 (1985). Smolensky, *Information processing in dynamical systems: foundations of harmony theory*, in *Parallel Distributed Processing*, MIT Press (1986).

11. Structure–function and brain criticality

The final chapter of the neural half of the course puts together three threads: the *structural* connectome (which neurons connect to which), the *functional* dynamics on top of it (Chapter 5), and the hypothesis — increasingly supported by data — that healthy brain activity operates near a critical point. This is where the ecological criticality of Chapter 3 and the neural criticality of Chapter 5 meet.

11.1. The brain criticality hypothesis

The observation is qualitative but robust: neural avalanches in vitro, LFP power spectra in vivo, fMRI resting-state fluctuations, and behavioural reaction-time distributions all show approximate **power-law** statistics, consistent with a system tuned close to a phase transition.

Definition 11.1 (Neural avalanche). A **neural avalanche** is a cluster of firing events separated from other clusters by silent intervals longer than a chosen threshold. Its *size* S is the number of events in the cluster; its *duration* T is the interval spanned.

Theorem 11.1 (Beggs–Plenz scaling). *In cortical slice cultures (rat somatosensory cortex, Beggs & Plenz, 2003) the size and duration distributions of neural avalanches follow*

$$P(S) \sim S^{-3/2}, \quad P(T) \sim T^{-2}, \quad \langle S \rangle(T) \sim T^2,$$

with exponents that are consistent with the mean-field directed percolation universality class. Corresponding measurements in vivo (monkey and human cortex) reproduce the same exponents to within a factor of two.

Remark. The mean-field DP class is what one expects from a *critical branching process*: each firing event triggers on average one descendant. Sub-critical values give fewer descendants and shorter avalanches; super-critical values give runaway activity (a seizure). The healthy brain sits, on this hypothesis, exactly at the critical point.

11.2. The Haldeman–Beggs (HTC) model

To connect avalanche statistics to structural connectomes, Haldeman & Beggs proposed a minimal *susceptible–active–refractory* three-state model on a lattice (or graph). Each node i is at any time in state $s_i \in \{S, A, R\}$; the update rules are:

Definition 11.2 (HTC update rules). Given the structural graph with adjacency W_{ij} and threshold θ , on each discrete time step:

- If $s_i = S$ and $\sum_j W_{ij} \mathbf{1}[s_j = A] \geq \theta$, then $s_i \rightarrow A$ (activation).
- If $s_i = S$ with sub-threshold input, then $s_i \rightarrow A$ with probability r_1 (spontaneous activation).

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- If $s_i = A$, then $s_i \rightarrow R$ (deterministic firing).
- If $s_i = R$, then $s_i \rightarrow S$ with probability r_2 (recovery).

The parameters (θ, r_1, r_2) set the operating point. The order parameter is the fraction of active nodes, and phase transitions in the model can be located analytically in the *continuous-time* mean-field limit.

Proposition 11.2 (HTC phase diagram). *The HTC model has three phases: sub-critical (activity decays), critical (power-law avalanches), and super-critical (spontaneous bulk activity). The critical line is $\theta = \theta_c(r_1, r_2)$, and at the transition the avalanche exponents match the empirical Beggs–Penz values.*

11.3. Connectomes from diffusion MRI and function from fMRI

Feeding the HTC model *real* structural connectomes rather than lattices is what makes the theory testable against neuroimaging data.

Example 11.1 (DTI–fMRI comparison)

Using diffusion-tensor imaging (DTI) to reconstruct a ~ 100 -region structural connectome and resting-state fMRI to record spontaneous activity time series, one can (i) simulate HTC dynamics on the DTI graph and (ii) compare the model *functional connectivity* C_{ij}^{model} (correlations of simulated node activities) with the empirical fMRI functional connectivity. The best fit occurs when the model operates at its critical point $\theta = \theta_c$, with model–data correlations of $r \approx 0.7$. Away from criticality the fit degrades sharply.

11.4. Continuous-time approximation

For analytical progress the HTC model is coarse-grained to a continuous-time density field $\rho(\mathbf{x}, t)$ satisfying a reaction–diffusion equation:

$$\partial_t \rho = D \nabla^2 \rho + a\rho - b\rho^2 - c\rho^3 + \sqrt{2\rho} \eta(\mathbf{x}, t),$$

where η is space-time white noise. This is the *directed-percolation* Langevin equation.

Theorem 11.3 (Fokker–Planck / Langevin equivalence for HTC). *The mean-field HTC dynamics reduces to the Langevin equation*

$$d\rho = (a\rho - b\rho^2) dt + \sqrt{2\rho/N} dW,$$

whose Fokker–Planck equation admits an exact stationary solution

$$P_{ss}(\rho) \propto \rho^{-1} \exp(-N \int^\rho \frac{a-b\rho'}{\rho'} d\rho').$$

Above criticality ($a > 0$) the mode of P_{ss} is at $\rho^ = a/b$; below criticality the distribution is concentrated near zero. At exactly $a = 0$ the drift is purely quadratic and the stationary distribution has a power-law form.*

11.5. Thermodynamic limit

For an infinite-size HTC model the fluctuations are washed out and one has a clean *second-order* phase transition with mean-field exponents.

Proposition 11.4 (HTC equilibria in the thermodynamic limit). *In the limit $N \rightarrow \infty$ the HTC order parameter ρ^* satisfies:*

- *Sub-critical ($a < 0$): $\rho^* = 0$.*
- *Critical ($a = 0$): $\rho^* = 0$ but with anomalously slow relaxation and long correlations.*
- *Super-critical ($a > 0$): $\rho^* = a/b > 0$.*

Linear stability of the equilibria is set by the sign of $a - 2b\rho^$; the sub-critical fixed point is stable for $a < 0$, the super-critical for $a > 0$.*

Corollary 11.5 (Critical relaxation and $1/f$ noise). *At criticality the relaxation time of the order parameter diverges as $\tau \sim (a - a_c)^{-1}$. The corresponding power spectrum of order-parameter fluctuations is a Lorentzian at frequency $1/\tau$; superposing many degrees of freedom with a distribution of relaxation times gives the $1/f$ noise observed in EEG.*

11.6. Stability analysis of equilibria

Small perturbations $\delta\rho$ around the mean-field equilibrium ρ^* satisfy

$$\partial_t \delta\rho = D\nabla^2 \delta\rho + (a - 2b\rho^*)\delta\rho + \sqrt{2\rho^*/N} \eta,$$

which is a stochastic diffusion equation with mass $|a - 2b\rho^*|$. The correlation length $\xi = \sqrt{D/|a - 2b\rho^*|}$ diverges at $\rho^* = a/(2b)$ — the critical point.

Example 11.2 (Super-critical vs sub-critical spectra)

The power spectrum of $\delta\rho$ at wavevector $k = 0$ is

$$S(\omega) = \frac{2\rho^*/N}{\omega^2 + (a - 2b\rho^*)^2},$$

a Lorentzian whose width shrinks to zero at the critical point. Empirical EEG spectra show precisely this critical slowing on approach to seizures.

11.7. Why would evolution build a critical brain?

Two arguments are often given, both quantitative:

- **Dynamic range.** A network responds to a range of input intensities that is *maximised* at criticality. Sub-critical networks saturate quickly; super-critical networks are unresponsive. Kinouchi & Copelli (2006) proved this analytically for a simple neural model.
- **Information transmission.** The mutual information between input and output patterns of a random neural network is maximal at the edge of chaos. This is a rigorous statement in reservoir computing (Bertschinger & Natschläger, 2004) and matches the empirical observation that cortical activity operates near $\sigma\sqrt{N} \approx 1$.

Remark (Cross-domain parallels). The mathematical statements — power-law avalanches with mean-field exponents, divergent correlation lengths, $1/f$ spectra — are literally the same as those we met in the neutral theory of biodiversity and in the SCS random rate network. Universality in the technical

sense of statistical mechanics is what allows insights to transfer from ecology to the brain.

11.8. Exercises

Exercise 11.1

Derive the mean-field HTC equilibrium $\rho^* = a/b$ from the coarse-grained Langevin equation. Show that the corresponding correlation length diverges as $|a - a_c|^{-1/2}$.

Solution

Set the deterministic drift to zero: $a\rho^* - b(\rho^*)^2 = 0$ gives $\rho^* = 0$ or $\rho^* = a/b$. Linearising around $\rho^* = a/b$ and looking for solutions $\delta\rho \sim e^{ikx - \omega t}$ gives dispersion $\omega = Dk^2 + (a - 2b\rho^*) = Dk^2 - a$. The correlation length is $\xi = \sqrt{D/a}$, which diverges as $a \rightarrow 0^+$ with exponent $\nu = 1/2$.

Exercise 11.2

Show that the mean-field DP exponents $\tau_S = 3/2$, $\tau_T = 2$, $\gamma = 2$ satisfy the scaling relation $\tau_S = 1 + (\tau_T - 1)/\gamma$. Comment on the physical meaning of the relation.

Solution

Substitute: $\tau_S = 1 + (2 - 1)/\gamma = 1 + 1/2 = 3/2$. The relation is equivalent to $\langle S \rangle(T) \sim T^\gamma$, which for a critical branching process means that avalanches with duration T have a *typical* size that scales as a power. Consistency of the exponents is the operational meaning of “the system is at a critical point”; any single exponent could be reproduced by accident, but the *relation* between them requires criticality.

11.9. Callouts

Note

Brain criticality is a *hypothesis*. It is well-supported for cortical avalanche statistics but not universally accepted for whole-brain fMRI. The reader should treat the numerical matches as suggestive rather than definitive.

Tip

When you fit a power law to neural data, always check that the range spans at least two decades and that the maximum-likelihood estimate is compared against an exponential null hypothesis using the Clauset–Shalizi–Newman test. A one-decade “power law” is not a power law.

Warning

The distinction between *criticality* and *quasi-criticality* (a system that is *not* fine-tuned but whose slow modes look critical) matters for the causal argument. Only genuinely critical systems have divergent susceptibilities.

 Caution

Interpreting $1/f$ EEG noise as evidence of brain criticality is contested. Many non-critical mechanisms (superposed Ornstein–Uhlenbeck processes, multiplicative noise) also produce $1/f$ spectra. A single spectral shape is not proof of criticality.

11.10. Sources

Prof. Suweis, *PMLS Course Notes*, Chapter 10. Original references: Beggs & Plenz, *Neuronal avalanches in neocortical circuits*, J. Neurosci. **23**, 11167 (2003); Haldeman & Beggs, *Critical branching captures activity in living neural networks and maximizes the number of metastable states*, Phys. Rev. Lett. **94**, 058101 (2005); Chialvo, *Emergent complex neural dynamics*, Nature Physics **6**, 744 (2010).

